

Leveraging Common Underlying Proficiency for Two-Stage Low-Resource MT Fine-Tuning

Anthony Baybayon
anthony_baybayon@dlsu.edu.ph
De La Salle University
Manila, Philippines

Jose Paolo Cruzado
jose_paolo_cruzado@dlsu.edu.ph
De La Salle University
Manila, Philippines

Ronald Dawson Catignas
dawson_catignas@dlsu.edu.ph
De La Salle University
Manila, Philippines

Mariel Tamondong
mariel_tamondong_a@dlsu.edu.ph
De La Salle University
Manila, Philippines

Abstract

This study examines the application of the Common Underlying Proficiency (CUP) theory to enhance machine translation for low-resource Philippine languages using a two-stage fine-tuning strategy. The NLLB-200-Distilled-600M model was adapted using LoRA-based parameter-efficient tuning to evaluate whether preliminary fine-tuning on a linguistically related low-resource language can improve performance on a target language. Experiments were conducted on Cebuano and Waray using Bible-domain parallel corpora and evaluated with BLEU and METEOR. The results indicate that, while single-stage fine-tuning significantly improves translation quality over the baseline, the CUP-inspired two-stage approach yields only marginal improvements within the training domain and reduced general-domain performance on FLORES-200. These findings suggest that linguistic similarity may offer limited positive transfer under constrained training conditions, and that domain specialization plays a major role in determining the generalizability of low-resource MT models.

CCS Concepts

• **Computing methodologies** → **Machine translation**; *Neural networks*.

Keywords

machine translation, low-resource languages, multilingual models, transfer learning, Common Underlying Proficiency, NLLB-200

1 Introduction

This paper aims to integrate one of the key multilingual theories, Common Underlying Proficiency (CUP) with fine-tuning large language models for machine translation. This theory states that at the surface level, the knowledge of multiple languages (L1 and L2) appear to be distinct and have no relation to each other. However, on the deeper level, both L1 and L2 are supported by a collection of common foundational knowledge. Furthermore, a development in either L1 or L2 makes this common foundational knowledge more vast, making it easier to learn or develop the other language [6]. Despite this, there exists a possibility of a negative transfer of knowledge, wherein a language rule, vocabulary or structure in L1 is incorrectly transferred to L2. This often happens if L1 and L2 are distantly related languages in terms of their components

(semantics, syntax, morphology) [4]. In the context of this paper, this theory will be leveraged by having two fine-tuning setups, one where this theory is applied and one where it is not considered. This dynamic between languages can extend to more than just two languages, which this study aims to leverage.

In the Philippines, over 100 languages are being spoken across different areas of the nation, with Filipino and English as the main languages [9]. In the NLLB-200 model, only Filipino and English are listed as high resource languages, while other Philippine languages are labeled as low resource languages. If CUP is proven to have a positive impact on fine-tuning languages, it can be leveraged to further improve machine translation of closely similar low resource languages.

2 Related Literature

2.1 Large Language Models for Translation

The development of Large Language Models (LLMs) has advanced the field of machine translation, especially through the creation of multilingual Transformer architectures. Early MT models laid the foundation for attention-based architectures, which revolutionized the traditional neural machine translation models [2, 12]. These architectures evolved further to multilingual systems, allowing the development of multilingual LLMs that demonstrate strong cross-lingual generalization. This led to the possibility of the transfer of knowledge from high-resource languages to low-resource ones, enabling the improvement of machine translation quality. [1, 7]

2.2 Linguistics and Transformer-based LLMs

Despite being trained with no explicit linguistic capability or knowledge, Transformer-based LLMs are able to mimic human linguistics. Recent research shows that Transformers encode hierarchical syntactic information despite the absence of explicit linguistic supervision during training. Modern analyses demonstrate how LLMs gain increasingly refined syntactic representations in newer Transformer architectures [8]. In parallel, large-scale surveys of Transformer interpretability confirm that these models learn layered linguistic features involving morphology, syntax, and semantics [10]. These findings show that Transformer LLMs generalize linguistic structure in ways that align with established linguistic theory. This phenomenon aligns with the possibility of applying

frameworks such as the Common Underlying Proficiency (CUP) theory on fine-tuning models.

3 Methodology

3.1 Data Collection

The study will use 2 low resource languages (Waray and Cebuano), and 1 high resource language (English). To build the multilingual corpora that are comparable, the data from bible texts in the 3 different languages is collected from a single source utilizing a custom web scraper. The scope of the corpora covers the entirety of the New Testament and some books of the Old Testament from the Bible. This also means that the domain of the dataset is heavily biased to the bible dataset.

Table 1: Raw Text Word Distribution

Language	Word Count
English	398,682
Cebuano	446,884
Waray	439,141

3.2 Preprocessing

After collection, all corpora were cleaned using a uniform preprocessing pipeline to prepare them for parallel corpus alignment and analysis. This process consisted of four main steps:

- (1) **Parse and Structure:** Raw verse data was parsed from text files using regular expression matching to extract book, chapter, verse numbers, and text content, organizing them into structured dictionaries keyed by verse references.
- (2) **Remove Leading Verse Numbers:** Leading numerical verse markers (e.g., "1 The book...") were removed from the beginning of text using pattern matching to isolate the actual verse content.
- (3) **Normalize Whitespace:** All consecutive whitespace characters (spaces, tabs, newlines) were normalized to single spaces, and leading/trailing whitespace was stripped.
- (4) **Validate and Filter:** Verses were validated to exclude empty entries and explicitly marked missing verses, ensuring only complete, valid text remained in the dataset.

Due to hardware constraints, the dataset is filtered further to eliminate verses with more than 128 characters. This is done to accommodate the maximum length in the training parameters. If the verses are not dropped, they get truncated which may result in inconsistencies and noise data.

Table 2: Supported and Dropped Verses Distribution

	Field Length	Verse Count
Supported	128	4543
Dropped	>128	11240

This process results in a cleaned version of the texts of 3 languages (English, Cebuano, and Waray) with aligned verse references. Following cleaning, parallel corpora were created for the language pairs English-Cebuano and English-Waray. Finally, each

parallel corpus was split into training, development, and test sets using a 80/10/10 ratio. A deterministic random split with seed 67 was applied using shuffled indices to ensure reproducibility while maintaining alignment across all language pairs.

This preprocessing pipeline results in a normalized, parallel corpus of 3 languages with consistent splits totaling 4543 verses per corpus, ready for tokenization, and machine translation fine-tuning.

3.3 Model Description

The model used in this study is Facebook/NLLB-200-Distilled-600M, a multilingual Sequence-to-Sequence model, developed as part of the No Language Left Behind (NLLB) project. The model uses a shared SentencePiece vocabulary for tokenization, enabling it to support 200 languages. Currently, this model has native support for Tagalog (tgl_Latn), Cebuano (ceb_Latn), and Waray (war_Latn). Among these, Tagalog is classified as a high-resource language, while Cebuano and Waray are classified as low-resource [5].

As for the architecture of the model, the NLLB-200-Distilled-600M model is a compact Transformer-based Sequence-to-Sequence system composed of a 12-layer encoder and a 12-layer decoder. Each layer follows the standard Transformer design, consisting of Multi-Head Self-Attention, a Feed-Forward Network (FFN), residual connections, and Layer Normalization. The encoder is responsible for contextually encoding the source sentence, while the decoder generates the translation autoregressively. The model operates with a 1024-dimensional embedding space, 16 attention heads per layer, and FFN layers with a hidden size of 4096. Since this distilled version does not use Mixture-of-Experts (MoE) layers, all computations pass through a dense FFN, making it significantly smaller and more efficient than the full 54B-parameter NLLB model [5].

3.4 Training Configuration

Training was conducted using the Hugging Face transformers library as the main framework. This is done in consideration of the compatibility for NLLB-200-Distilled-600M. The precision for the tensors is set to Brain Float 16 (BF16) to optimize training time and memory usage without sacrificing the quality of the model.

To further optimize the training process, Low Rank Adaptation (LoRA) is implemented. This framework lessens the parameters needed to be trained for downstream tasks (i.e. fine-tuning). Despite lessening the trainable parameters, LoRA does not present worse performance in terms of quality when compared to full fine-tuning (Hu et al., 2021). However, LoRA is fundamentally different from full fine-tuning, and cannot be treated as its equivalent (Shuttleworth et al., 2024). The implementation of LoRA for the fine-tuning of NLLB-200 targeted modules q_proj , v_proj , k_proj , and o_proj as these are the key components of the attention layer of LLMs. As for the specific LoRA parameters, rank is set to 16, alpha is set to 32, and dropout rate is set to 0.1 to optimize performance and memory usage. The epoch is set to 6, as to avoid overfitting of the dataset. In line with this, the learning rate is set to 1×10^{-5} .

3.5 Low Resource Language Similarity Index

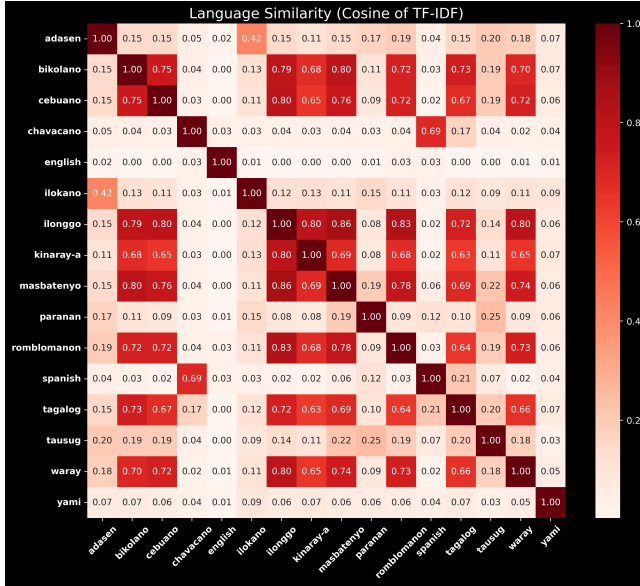


Figure 1: Similarity Matrix of the 16 Languages

To quantify the linguistic similarity between the languages, we calculated the cosine similarity of character trigrams ($n = 3$) derived from their respective corpora. Prior to computation, we balanced the trigram features by performing a Term Frequency-Inverse Document Frequency (TF-IDF) transformation on the raw frequencies to downweight common, non-unique patterns. Cosine similarity measures the angle between two vectors in a multi-dimensional space, evaluating the directional alignment of n -gram profiles regardless of the magnitude or total size of the corpora.

Applying this methodology to our target languages revealed a high similarity score of 0.72 between Cebuano and Waray (Figure 1). This quantitative evidence confirms significant orthographic overlap, providing the empirical basis for the Common Underlying Proficiency (CUP) experiment detailed in the following section.

3.6 Control and Experimental Setups

In this study, the main objective is to apply the concept of Underlying Proficiency in fine-tuning LLMs. With that in mind, there will be a comparison of the performance of a model that is fine-tuned directly to the low resource language with a model that is fine-tuned in a similar low resource language first before the actual target. This comparison will test if the application of Common Underlying Proficiency yields better translation performance.

For the control setups, a direct fine-tuning of English to Waray and English to Cebuano translation is done. This setup serves as the baseline or conventional training approach for fine-tuning models. For the experimental setup, a 2-stage approach is done. First, the model is trained in translating English to a similar language. In the context of this study, the similar language pairs are Cebuano and Waray, based on their similarity index of 0.72. Next, the fine-tuned model is trained in translating English to the target language

texts. The 2-stage approach is inspired by the CUP. As Waray and Cebuano share a fairly high similarity index, there might be a possibility to leverage learning Cebuano before learning Waray, and vis-a-vis.

3.7 Evaluation

The quality of the translation will be measured through Bilingual Evaluation Understudy (BLEU) and Metric for Evaluation of Translation with Explicit Ordering (METEOR). BLEU is a measure of how similar the n -grams or word sequences of the machine translation are to a reference translation. On the other hand, METEOR is an aggregate measure of precision, recall and penalty of the translations [3, 11].

With this in mind, the models generated by both setup will be tested on two different datasets, the Bible dataset that have been scraped, and the Facebook Low Resource Language Evaluation Set (FLORES-200) *devtest* dataset, which was used by the NLLB200 for benchmarking. The FLORES-200 *devtest* is a human-curated validation dataset with a generalized domain mined from various sources such as Wikipedia articles [5].

4 Results and Interpretation

Table 3: Evaluation with Bible Dataset for Cebuano

Model	BLEU	METEOR
NLLB-200-600M (Pre-Trained)	26.27	0.5415
NLLB-200-600M (Control-Baseline)	29.22	0.5735
NLLB-200-600M (Experimental-Stage 2)	29.42	0.5744

Table 4: Evaluation with Bible Dataset for Waray

Model	BLEU	METEOR
NLLB-200-600M (Pre-Trained)	20.93	0.4567
NLLB-200-600M (Control-Baseline)	27.90	0.5264
NLLB-200-600M (Experimental-Stage 2)	28.20	0.5379

Tables 3 and 4 present the BLEU and METEOR scores of the baseline pre-trained NLLB-200 model, the direct fine-tuning setup (Control), and the 2-stage CUP-inspired fine-tuning (Experimental Stage 2) for Cebuano and Waray.

Across both languages, fine-tuning provides a clear improvement over the base model. This confirms that adaptation to domain-specific bilingual corpora strengthens translation quality even when using a distilled 600M-parameter model. For Cebuano, direct fine-tuning increased BLEU from 26.27 to 29.22, while the 2-stage CUP setup slightly further raised it to 29.42. A similar trend is observed for Waray, where BLEU increased from 20.93 (Pre-trained) to 27.90 (Control), and again to 28.20 (Experimental). METEOR scores follow the same trend of scores.

The margins of improvement, however, are noticeably small between the Control and Experimental setups. While fine-tuning on a related low-resource language first (e.g., Cebuano $\rightarrow$ Waray or Waray $\rightarrow$ Cebuano) does contribute a measurable gain, the

magnitude of this gain is modest. In both language pairs, the 2-stage approach improved BLEU by only +0.20 for Cebuano and +0.30 for Waray. These increments fall within what is typically considered noise-level variations in MT evaluation, which suggests that the benefit from CUP-aligned initialization is limited under the given conditions.

This finding indicates that while linguistic similarity and positive transfer exist conceptually, their practical impact on NLLB-style models is constrained—possibly due to the model’s pre-existing cross-lingual alignment, the small training corpus, and domain specificity. Overall, the CUP-inspired 2-stage approach offers marginal improvements in a controlled, Bible-domain setting, but does not significantly outperform single-stage fine-tuning.

Table 5: Evaluation with FLORES-200 Dataset for Cebuano

Model	BLEU	METEOR
NLLB-200-600M (Pre-Trained)	28.62	0.5560
NLLB-200-600M (Control)	28.25	0.5565
NLLB-200-600M (Experimental Stage 2)	27.65	0.5500

Table 6: Evaluation with FLORES-200 Dataset for Waray

Model	BLEU	METEOR
NLLB-200-600M (Pre-Trained)	28.17	0.5491
NLLB-200-600M (Control)	26.10	0.5279
NLLB-200-600M (Experimental Stage 2)	23.98	0.4995

The FLORES-200 evaluation provides a different perspective on model behavior. As shown in Tables 5 and 6, both fine-tuning setups perform worse than the original pre-trained model. For Cebuano, the experimental 2-stage setup drops from a BLEU of 28.62 (Base) to 27.65, and for Waray the drop is more severe, falling from 28.17 to 23.98. METEOR decreases in parallel.

This degradation strongly supports the conclusion that the models overfit heavily to the Bible domain. As a result, the model becomes more specialized for Bible translations but loses its general-purpose capability when evaluated on the much broader FLORES-200 dataset, which contains generalized domain content.

Furthermore, the 2-stage CUP approach performs worse than the direct fine-tuning on FLORES-200. This suggests that the additional fine-tuning stage reinforces the domain bias twice: first on Language A Biblical text, then on Language B Biblical text. This narrows the representation space even further. Thus, the model moves farther away from general purpose translation, resulting in lower metric scores.

The difference in FLORES behavior demonstrates that positive transfer in low-resource languages is domain-sensitive: while the Bible-aligned corpora enable improvements within the same domain, they do not generalize beyond it. This also clarifies why the CUP-inspired approach did not yield significant benefits, it amplifies domain adaptation rather than enabling broad cross-lingual generalization.

5 Conclusion and Limitations

5.1 Conclusion

This study explored whether the Common Underlying Proficiency (CUP) theory, originally proposed to describe human multilingual learning, could be effectively applied to the fine-tuning process of large multilingual machine translation models, specifically NLLB-200-Distilled-600M.

The results indicate that while the model benefits from domain-specific fine-tuning, the direct application of CUP through a 2-stage training process does not yield substantial performance gains. Although the experimental setup consistently outperformed the direct fine-tuning baseline within the Bible domain, the improvements were minimal and did not reach thresholds typically considered significant in MT research.

More importantly, when tested on the FLORES-200 devtest dataset, both the baseline and the CUP-based fine-tuned models performed worse than the original model, confirming overfitting to the narrow Bible domain. The CUP-based model performed even worse in this out-of-domain evaluation, suggesting that the additional training stage further limited the model’s linguistic generality.

These findings highlight a distinction between human and machine learning processes. In humans, CUP enables shared cognitive structures that facilitate cross-lingual knowledge transfer. While in LLMs, cross-lingual alignment already exists by architectural and training design, and additional fine-tuning on highly similar languages does not necessarily strengthen this alignment, especially when data is biased to a certain domain.

Thus, this study concludes that the CUP theory, while conceptually related to multilingual representation learning, does not translate into meaningful gains in current Transformer-based MT fine-tuning for these Philippine low-resource languages under the constraints tested.

5.2 Limitations

In line with results, this project has several limitations that should be acknowledged:

- (1) **Corpus Domain:** The most significant limitation is the nature of our corpus. All corpora were sourced exclusively from Bible texts. This is a single, highly specialized domain that does not capture the full breadth of vocabulary used in modern, everyday language. This specialization may negatively impact the model’s performance when translating general, out-of-domain text.
- (2) **Hardware Capabilities:** Another limitation is the computational constraints of our hardware, specifically the GPU. This restricted the scope of the models for training.
- (3) **Lack of Human Evaluation:** It is not enough that the only metric measured and evaluated is BLEU. A human evaluation of translated texts would give more specific evaluation, especially when it comes to semantics.

Future work could address these limitations by incorporating more diverse corpora from multiple domains and by utilizing cloud-based high-performance computing resources. Furthermore, metrics aside from BLEU could also be incorporated into future studies in relation to machine translation.

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